

AOD Addition Pattern Standardisation

Role

Operations Analyst

Company

Jindal Stainless Ltd

Department

Steel Melting Shop, AOD Operations

1. Project Overview

This sample work is based on the standardisation of the AOD addition pattern in the Steel Melting Shop. The objective of the work was to improve process consistency, reduce process time, and support productivity improvement during AOD operations.

The AOD process involves the controlled addition of Dolomite, Lime, HCFEcr, FeNi, FeMn, and flux at different stages of the blowing process. If the addition sequence is not followed correctly, it can create variation in process time, shift performance, material control, and operational discipline.

The work focused on converting shop-floor operating practice into a structured Standard Operating Procedure that could be followed by Shift-In-Charges and operators during production.

2. Business Problem

The AOD addition process was dependent on shift-level execution and operator experience. Since multiple materials were added at different stages of the blowing process, there was a need to clearly define the correct sequence, timing, responsibility, and exception rules.

Key problems identified were:

Area	Problem
Process consistency	Different shifts could follow slightly different addition practices
Process time	Variation in addition sequence could increase AOD process time
Material control	Flux and alloy addition needed clear separation
Shift responsibility	Shift-In-Charges needed a clear reference document
Operational risk	Incorrect addition timing could affect process stability
Training	New or less experienced team members needed standard guidance

3. Objective

The objective was to create a clear and standardised AOD addition pattern for production teams.

The main objectives were:

Objective	Description
Reduce process variation	Standardise the addition pattern across shifts
Improve productivity	Support reduction in process time through a controlled sequence
Improve process control	Define when flux, alloys, Lime, and Dolomite should be added
Improve shift clarity	Provide Shift-In-Charges with a structured operating reference
Reduce execution dependency	Move from experience-based practice to documented process
Improve compliance	Ensure the same process is followed during AOD operation

4. Scope

The scope of the SOP covered the addition pattern followed during AOD operations.

The scope included:

Process Area	Included in Scope
<i>Zero batch addition</i>	Dolomite and Lime addition before pouring
<i>Blow start addition</i>	Dolomite addition after blow starts
<i>Alloy addition</i>	HCFeCr, FeNi, and FeMn addition
<i>Flux control</i>	Timing and restriction of flux addition
<i>Main blow</i>	Lime and flux addition till end of main blow
<i>Second and third blow</i>	Remaining flux addition
<i>Exception handling</i>	High opening silicon percentage and grade sequence rules
<i>Safety</i>	PPE requirements for shop-floor execution

5. Stakeholders

STAKEHOLDER	RESPONSIBILITY
SHIFT-IN-CHARGE	Responsible for following and controlling the SOP during the shift
AOD OPERATORS	Execute material addition activities as per instruction
PRODUCTION MANAGER	Monitor productivity, process time, and shift performance
QUALITY TEAM	Review the impact of process consistency on product quality
SAFETY TEAM	Ensure PPE and safe working practices are followed
MAINTENANCE TEAM	Support equipment availability and smooth operation

6. As-Is Process

Before standardisation, the AOD addition pattern was mainly followed through practical experience and shift-level understanding.

Area	As-Is Situation
<i>Process documentation</i>	Addition sequence was not clearly structured in one document
<i>Execution method</i>	Operators followed instructions based on experience and shift practice
<i>Shift variation</i>	Different shifts could have minor differences in addition timing
<i>Flux control</i>	Flux addition needed clear restriction during alloy addition
<i>Responsibility</i>	Shift-In-Charge responsibility existed but needed formal documentation
<i>Training</i>	New team members depended on verbal instructions and observation

7. To-Be Process

The To-Be process introduced a clear and standard addition sequence for AOD operations.

Step	Standardised Activity
1	Add zero batch with Dololime and Lime before pouring
2	Add Dololime when the blow starts
3	Add HCFeCr slowly before the defined oxygen requirement
4	Do not add flux while HCFe alloy addition is ongoing
5	Add Dololime flux after HCFe alloy addition is completed
6	Add FeNi and FeMn as required
7	Do not add flux during FeNi and FeMn addition
8	Add Lime till the end of main blow after alloy addition is completed
9	Add remaining flux during second and third blow
10	Apply exception rules for high opening silicon percentage and grade sequence

8. Business Rules

Rule ID	Business Rule
BR-01	Zero batch must include the defined Dololime and Lime quantity
BR-02	Dololime must be added when the blow starts
BR-03	HCFeCr must be added slowly before the defined oxygen requirement
BR-04	Flux must not be added while HCFe alloy addition is ongoing
BR-05	Dololime flux must be added only after HCFe alloy addition is completed
BR-06	FeNi and FeMn addition must not overlap with flux addition
BR-07	Lime must be added till the end of main blow after alloy addition is completed
BR-08	Remaining flux must be added during second and third blow
BR-09	If opening silicon percentage is high, flux quantity must be increased as per process requirement
BR-10	Charging hopper selection must be fixed before starting any grade sequence
BR-11	For specific grade sequences, additional process precautions must be followed to protect AOD life

9. Safety Requirements

The SOP included basic PPE requirements for safe shop-floor execution.

PPE	Purpose
<i>Helmet</i>	Head protection
<i>Safety shoes</i>	Foot protection in shop-floor area
<i>Nose mask</i>	Dust and fume protection
<i>Hand gloves</i>	Hand protection during material handling
<i>Jacket</i>	Body protection near operating area

10. Process Controls

The SOP introduced the following process controls:

Control Area	Control Description
Material sequencing	Defined when each material should be added
Flux restriction	Prevented flux addition during alloy addition
Shift responsibility	Assigned execution responsibility to Shift-In-Charges
Exception handling	Defined conditions for high opening silicon percentage
Grade sequence control	Required charging hopper selection before grade sequence
Safety compliance	Listed PPE requirements for execution

11. Operations Analyst View

This work involved analysing the existing production process and converting it into a controlled operating method.

Operations analysis activities included:

Activity	Description
Process observation	Studied how material addition was performed during AOD operation
Gap identification	Identified areas where sequence variation could occur
Standardisation	Created a single operating method for all shifts
Productivity focus	Linked addition pattern control with process time reduction
Risk control	Reduced chances of incorrect or overlapping material addition
Compliance support	Created a documented reference for shift-level execution

12. Business Analyst View

This work also matched Business Analyst responsibilities because it required understanding the process, identifying rules, documenting requirements, and converting operational knowledge into a usable document.

BA Skill	Application in This Work
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Requirement gathering	Captured process requirements from AOD operations
Process mapping	Converted shop-floor activities into a step-by-step process
Business rules documentation	Documented material addition rules and exception conditions
Stakeholder coordination	Worked with production and shift-level stakeholders
As-Is analysis	Reviewed the existing addition process
To-Be design	Defined the standardised future process
Documentation	Prepared structured SOP content
Risk analysis	Identified risk of wrong sequence and overlapping additions
Operational improvement	Connected process control with productivity improvement

13. Expected Benefits

AREA	EXPECTED BENEFIT
PROCESS CONSISTENCY	Same addition sequence followed across shifts
PRODUCTIVITY	Reduced process variation and better control over process time
MATERIAL CONTROL	Clear separation between flux and alloy addition
SHIFT EXECUTION	Better clarity for Shift-In-Charges and operators
TRAINING	Easier onboarding for new or less experienced team members
SAFETY	Clear PPE requirements documented
COMPLIANCE	SOP available as a controlled operating reference

14. KPI Tracking

KPI	BEFORE STANDARDISATION	AFTER STANDARDISATION
AOD PROCESS TIME	Variable between shifts	More controlled and consistent
ADDITION SEQUENCE	Experience-based	SOP-driven
FLUX AND ALLOY CONTROL	Risk of overlap	Clearly separated
SHIFT COMPLIANCE	Dependent on individual practice	Based on documented SOP
OPERATOR GUIDANCE	Verbal or experience-based	Documented step-by-step instruction
PROCESS VARIATION	Higher chance of variation	Reduced variation

15. Final Summary

The AOD Addition Pattern Standardisation work focused on improving process consistency, reducing process time variation, and creating a structured operating method for shop-floor execution.

The work involved analysing the AOD production workflow, understanding material addition requirements, documenting the correct sequence, defining shift-level responsibility, capturing business rules, and including safety requirements.

This sample demonstrates practical experience in operations analysis, process standardisation, SOP documentation, business rules writing, stakeholder coordination, and productivity-focused process improvement in a manufacturing environment.